Emotion Detection of Thai Elderly Facial Expressions using Hybrid Object Detection

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Abstract— **An elderly population is a special group that needs to be taken care of closely. A key area of concern for the elderly is that of mental health and many technologies can be applied in this area. One possible tool is facial expression recognition (FER) that can be used to detect emotions of the elderly for the purpose of mental health care. In this research, we propose a hybrid of Faster R-CNN, SSD, and YOLOv5 object detection models for elderly facial expression detection. In our experiments, the proposed hybrid model was trained on a Thai elderly facial emotion dataset, and its performance was compared to a single-model of Faster R-CNN, SSD, and YOLOv5. The experimental results indicated that the proposed hybrid object detection model had achieved the best performance with an accuracy of 94.07%. This was comparatively better than YOLOv5, which gave the accuracyof 93.33%.**

Keywords- Facial Emotion Detection, Thai Elderly, Artificial Intelligence, YOLOv5

I. INTRODUCTION

An aging population has been increasing in many countries around the world because the birth rate is declining, and more advanced medical developments have allowed people to live longer. Thailand has become one of countries that is entering the agingsociety in 2022 [1]. Because of the decrease in the working age population, all countries that are entering an aging societyneed to develop new tools and technologies that can take carethe aging population autonomously. Many older people often have special needs and require help, especially with mental health because the mental states of some older adults that have mood or hyperactive disorders could lead to mental disorderswhich obstruct their daily living and quality of life [2].

Facial Emotion Recognition (FER) [3] is a technology that can be a key component to help solve the problem of taking care of the elderly at home. The automatic detection of emotionalexpression in older adults is important in mental health assessment. The elderly may live alone or have to be alone during the day. This can cause certain emotional states that caregivers may not be aware of. Therefore, FER has an important role in shaping the elderly's emotional detection system.

Convolution Neural Network (CNN) is a deep learning neural network used for image analysis. The CNN is popularfor its aggregate image analysis and can be used to create facial emotion recognition systems. Nowadays, there are many models based on the CNN. The popular models are SSD [4], Faster RCNN [5], YOLO [6], and RetinaNet [7] which yields better performance than others.

This research proposes a hybrid object detection model for detecting the facial emotions of the elderly that were trained on the facial expression image dataset of Thai elderly. We collected facial emotion image data of Thai elderly from online media, including dramas or variety shows with the elderly participating in the programs. All images in the dataset were then passed to evaluators to label the images and were used in the model training and testing processes. In this work, three models, Faster RCNN, SSD, and YOLOv5 [8], were combined into a hybrid object detection model and trained and tested on the Thai elderly facial emotion image dataset.

The rest of the paper is organized as follows. Section 2 is the content of research-related works to this research. Section 3 presents all the experimental steps. Section 4 describes the model's performance evaluation results. Section 5 is a summary of the content of this research.

II. RELATED WORKS

Y. Jang et al. [9] presented a novel single-shot face-related task analysis method, known as Face-SSD. It was capable of face detection, smile recognition, face attribute prediction, and valence-arousal estimation in the wild. The Face-SSD used a Fully Convolutional Neural Network (FCNN) for different size detection and face recognition. The Face-SSD was a network of non-preprocessing and FCNN architecture. Tests showed real-time detection performance was achieved at 21 FPS. Smile detection accuracy was 95.76% and attributes prediction accuracy was 90.29%.

A. F. Yaseen et al. [10] developed a hybrid model of facial expressions based on deep learning algorithms, which were EfficientNetB0, DenseNet169, and the model integration EfficientNetB0+VGG16. The evaluation used FER2013 and JAFFE datasets to measure the performance of the model. The result of the hybrid model of EfficientNetB0+VGG16 achieved 90.6% and 95.6% accuracy on the FER2013 and JAFFE datasets.

Qingqing Xu et al. [11] designed a face detector on YOLOv5 to create a new two-level face detection model called SR-YOLOv5 to solve the problem of detecting small faces in dense areas, by improving the backbone and loss function of YOLOV5. The research focused on improving mean average precision (mAP) and improving detection in blur areas or low resolution. Performance tests were performed on a wider face dataset, where levels were divided into easy, medium, and hard. The SR-YOLOv5 achieved mAP efficiency of 96.3% for easy, 94.9% for medium, and 88.2% for hard level.

Guan-Chun Luh et al. [12] studied facial emotion detection with YOLOv3. Three datasets, JAFFE, RaFD, and CK, were used to assess the model's effectiveness. The experiment showed that the detection performance of YOLOv3 was very good. The JAFFE database showed an accuracy of 98.12%; the RaFD database showed an accuracy of 97.01%, and the CK+ dataset showed an accuracy of 99.72%

D. Fan et al. [13] used an ensemble technique with CNN base model to optimize recognition rate. The sub-network models used in the ensemble model were VGG19 and ResNet18 which had been optimized together. Performance tests were done using the FER2013 dataset. The ensemble model's accuracy was 73.14%

C. Jia et al. [14] proposed a facial expression recognition based on CNN ensemble learning. This work combined the sub-networks including AlexNet, VGGNet, and ResNet into one model and used the SVM Classifier to combine the results. The FER2013 dataset was used to test the performance of this model. The resulting precision obtained from the tests of the ensemble model was 71.27%, while each sub-network, AlexNet, VGGNet, and ResNet, achieved 65.75%, 67.82%, and 67.96% accuracy, respectively.

Sergio Pulido-Castro et al. [15] studied the creation of real-time facial emotion detection for the perception of emotions in children with ASD. The model was designed as an ensemble of machine learning models. With this design approach, they were able to improve recognition to increase efficiency. The algorithm was tested with a tool to stimulate the imitation and perception of emotions of children with ASD based on their facial expressions. The accuracy value was 75.4%.

As can be seen from [10, 13, 14] research, using an ensemble model of sub-networks has consistently led to improved accuracy over using individual models. Hence, this workalso used this approach to compare the performance of the popular Faster RCNN, SSD, and YOLOv5 against that of a hybrid detection model combined from the three models.

III. METHODOLOGY

There are five steps for building the ensemble model. The first step is Data Collection. The second step is Image Preprocessing. The third step is Data Augmentation. The fourth step is Model Training, and the final step is to combine all models (Faster RCNN, SSD, and YOLOV5) to build the hybrid object detection model.

A. Data Collection

The dataset used in our experiment comprised of facial expression images of Thai elderly people which were collected from online media of various social platforms, including dramas and variety shows that have elderly people participating in the programs. Only the video clips that clearly showed the facial expressions of the elderly were used, and an image was collected from every 5 frames of the video clips for the dataset. There were 6 emotions in this dataset, namely, Angry, Happy, Neutral, Sad, Fear, and Surprised. About 200 images were initially collected for each emotion, resulting in a total of 1200 images. The dataset was then provided to 5 (human) evaluators to categorize the facial expression in each image based on their individual analysis. The class label of each image in the dataset was determined based on a majority voting criteria from the five votes. Images that did not receive the full five votes were excluded from the dataset. The image collection and labelling processes were repeated a few times to generate the finalized dataset with a total of 900 images, comprising of 150 images of each emotion. Example images with their corresponding voted emotions of the dataset are shown in Figure 1.

Figure 1. Example images of Thai elderly facial emotions.

B. Image Preprocessing

All images in the dataset were resized to a standard size for training and testing the model. The standard size for all models was 640x640 pixels. After resizing the image, the dataset was split into train, validate, and test sets. The splitting ratios were set to 40, 30, and 30, respectively.

C. Data Augmentation

In this process, two techniques were applied to increase the amount of data in the training set to improve the efficiency of all models. The first technique was cropping which zoomed the images at 11%, and the second technique was rotating that rotated the images in degrees of -7 to 7 [16, 17]. After performing the data augmentation with these techniques, the number of images in the training set increased to 1,080 images.

D. Base Model Training

As shown in Figure 2, the three models used in ourtraining process were SSD ResNet101 V1 FPN 640x640, Faster R-CNN ResNet101 V1 640x640, and YOLOv5m. All three models were trained with the same dataset and environment after the Image Preprocessing and Data Augmentation steps.

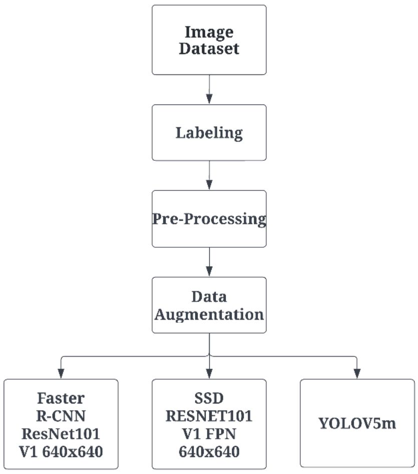

Figure 2. Training process of the three base models.

E. Hybrid Object Detection Model

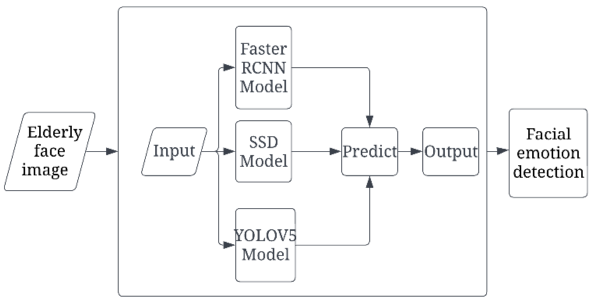

Figure 3. Training process of our hybrid model.

Our proposed hybrid object detection model combined three models into a single model. The proposed hybrid modelis shown in Figure 3. The proposed model would receive images of the elderly faces as input and feed them into the three base models. Then, each base model predicts the emotion of the elderly face image. This results in three predictions from the three base models, and these predictionsare voted to produce the output prediction of the hybrid model, using majority voting along with the following criteria:

1) The class label that gets two or more votes are considered as the output of the hybrid model.

2) If all models give different predictions, the output of thehybrid model will be selected from the prediction with the highest detection score.

IV. RESULT

After completing the training process. We use the three base models as well as the hybrid model to test the performance of evaluating facial emotion detections of Thai elderly with a test set of 270 images. The examples of the facial emotion detections are shown in Figure 4.

Figure 4. Examples of detected facial emotions with receiving scores.

Evaluation results of the Faster R-CNN model's performance in detecting facial emotions of the Thai elderly are shown in Table I. Happy are the most accurate predictive classes, with an F1 Score of 1 across all six classes. While Angry F1 Score is 0.88, which is the lowest.

TABLE I. EVALUATION OF FASTER R-CNN MODEL

Class	Precision	Recall	F1 Score
Angry	0.87	0.89	0.88
Fear	0.91	0.93	0.92
Happy	1	1	1
Neutral	0.98	0.91	0.94
Sad	0.92	0.98	0.95
Surprised	0.95	0.91	0.93

For the SSD model, the facial emotion detection results are shown in Table II. Happy are the most accurate detection classes, with classes having an F1 score of 1, and the class with the lowest F1 score is Angry with 0.87.

TABLE II. EVALUATION OF SSD MODEL

Class	Precision	Recall	F1 Score
Angry	0.87	0.87	0.87
Fear	0.96	0.96	0.96
Happy	1	1	1
Neutral	0.91	0.93	0.92
Sad	0.88	0.93	0.90
Surprised	0.98	0.89	0.93

The results of YOLOv5 model are shown in Table III. Happy and Fear are the best class with an F1 score of 0.97, while Angry and Neutral have an F1 Score of 0.91, which is the lowest.

TABLE III. EVALUATION OF YOLOV5 MODEL

Class	Precision	Recall	F1 Score
Angry	0.86	0.96	0.91
Fear	0.98	0.96	0.97
Happy	0.98	0.96	0.97
Neutral	0.93	0.89	0.91
Sad	0.9	0.96	0.92
Surprised	0.98	0.89	0.93

Lastly, the results of our hybrid model are shown in Table IV. The best class of hybrid model is Happy with an F1 Score of 1, followed by fear with an F1 value of 0.97. The lowest value is Angry with an F1 Score of 0.88.

TABLE IV. EVALUATION OF HYBRID MODEL

Class	Precision	Recall	F1 Score
Angry	0.87	0.89	0.88
Fear	0.98	0.96	0.97
Happy	1	1	1
Neutral	0.98	0.91	0.94
Sad	0.86	0.98	0.92
Surprised	0.98	0.91	0.94

Overall performance of each model is given in Table V. In summary, accuracy of theYOLOv5 is 93.33%; SSD accuracy is 92.96 %, and Faster R-CNN accuracy is 93.70%,whereas the hybrid model could achieve an accuracy of 94.07%. All results showed that the most detected emotion accuracy to the three models was Happy because Happy was facial feature that was different from all 5 facial expressions and was the most prominent in both the eyes and mouth shape. The least accurate of detecting emotion was Angry because Angry has somewhat similar facial features to other emotion expressions, e.g., Surprised, Sad, and Scared. This made detecting Angry more confused.

TABLE V. COMPARATIVE EVALUATION SUMMARY OF ALL MODELS

Model	Precision	Recall	Accuracy
Hybrid	0.94	0.94	94.07 %
Faster R-CNN	0.93	0.93	93.70 %
SSD	0.93	0.92	92.96 %
YOLOv5	0.93	0.93	93.33 %

V. CONCLUSION

This research focused on building a hybrid object detection model, based on the Faster R-CNN, SSD, and YOLOv5, for solving the emotion detection problems. The proposed model was applied to detect facial emotions of Thai elderly people. The dataset was collected from online social media having Thai elderly people. The dataset consisted of 900 images to train the model. After the three base models completed the training, they were combined to form a hybrid model, resulting in a total of four models. All models were tested on thesame test set. From the experimental results, the proposed hybrid model achieved the best accuracy of 94.07%, whereasthe YOLOv5 gave the accuracy of 93.33%, SSD with the accuracy of 92.96%, and Faster R-CNN with the accuracy of 93.70%. The hybrid model performed better than the others because the hybrid model was a combination of all three with ensembled voting method. This made hybrid models have more performance than single models. This result was consistent with related works previously discussed [10, 13, 14] in that the hybrid model had higher prediction performance.

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